Samuel Shilling

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Profile

Quantitative researcher with expertise in developing Python-based data processing pipelines, modelling complex datasets using Bayesian inference, and building simulations to predict future results. A proven track record in two peer-reviewed first-author publications in the Monthly Notices of the Royal Astronomical Society, a world-leading astrophysical research journal, with a third in preparation, and presentations at five research conferences. Seeking to apply innovative scientific methods to solve challenging problems in industry.

Skills

- **Programming and tools:** Python, Linux, Git, NumPy, pandas, SciPy, Matplotlib, Ollama (local LLM integration), APScheduler.
- **Statistical analysis:** Data cleaning and visualisation, MCMC simulations, Spearman's rank and partial correlation tests, bootstrapping, Kolmogorov–Smirnov tests, bias tests.
- **Data modelling:** Bayesian inference, nested sampling, posterior estimation, evidence-based model comparison, uncertainty quantification, non-linear least squares, linear regression, model extrapolation.

Experience

Doctoral Researcher, Astrophysics **Oct 2023 – Feb 2027 (expected)**
Lancaster University, UK

- Discovered a statistically significant correlation in a new data domain, ruled out sample and selection biases, and quantified its properties.
- Built a Python pipeline using CPU multiprocessing to model a large sample of complex astrophysical sources using Bayesian inference for model comparison and parameter estimation. Developed a simulation to test the inference.
- Presented research at five conferences (including three major national conferences) to expert and non-expert audiences; assisted supervising four MSc Astrophysics student research projects and teaching undergraduate physics courses; chaired the University Astrophysics journal club in rotation; led teams on NASA and ESA proposals.
- Developed a Python application (available on GitHub) that fully automates journal club role allocation indefinitely. This includes a locally hosted natural-language (Ollama) interface, so members can check and modify their roles in plain English, and a scheduled daily workflow.

Research Assistant, Astrophysics **Jul 2024 – Dec 2024**
NASA Goddard Space Flight Center, MD, USA

- Built a Python pipeline to collate and process 20 years of archival GRB data observed with NASA's *Swift* satellite.
- Discovered data quality (rather than the physics) was a major driver of uncertainty in previous published results. Built a simulation to predict the results in future datasets.
- Analysed newly detected transient astrophysical sources from NASA's *Swift* satellite, in real time (within minutes to hours) producing 37 community bulletin reports.

Education

PhD, Astrophysics **Oct 2023 – Feb 2027 (expected)**
Lancaster University, UK

- Thesis: "Statistical Analyses and Bayesian Inference in Populations of Gamma-Ray Burst Afterglows".

MPhys, Physics and Astrophysics **Sep 2018 – Jun 2023**
University of Sheffield, UK

- Grade: Class one honours (78% final year, 80% dissertation, 77% literature review).

First-author peer-reviewed publications

- Shilling et al. (in prep.) *Inferring GRB jet structures, physical parameters, and geometric properties from a large sample of X-ray afterglows.* In preparation (est. 2027).
- Shilling et al. (2026) *The intrinsic luminosity–decay correlation in subsamples of GRB X-ray afterglows.* MNRAS, 552, stag1589
- Shilling et al. (2025) *Evidence for an intrinsic luminosity–decay correlation in GRB radio afterglows.* MNRAS, 542, 2421